

Ex 11

$$1) X^{11} + X^{10} + X^9 = X^9(X^2 + X + 1)$$

2^{ème} méthode

$$\begin{array}{r|l} X^{11} + X^{10} + X^9 & X^2 + X + 1 \\ \hline X^{11} + X^{10} + X^9 & X^9 \\ \hline 0 & \end{array}$$

$$2) P = X^4 + X^3 + aX^2 + bX + 2 = (X^2 + 2)Q$$

$$\Rightarrow P(\sqrt{2}i) = 0$$

$$\Rightarrow (\sqrt{2}i)^4 + (\sqrt{2}i)^3 + a(\sqrt{2}i)^2 + b\sqrt{2}i + 2 = 0$$

$$4 - 2\sqrt{2}i - 2a + b\sqrt{2}i + 2 = 0$$

$$(6 - 2a) + (b - 2)\sqrt{2}i = 0$$

$$\Rightarrow \begin{cases} 6 - 2a = 0 \\ b - 2 = 0 \end{cases} \Leftrightarrow \begin{cases} a = 3 \\ b = 2 \end{cases}$$

$$3) P = (X+1)Q_1 + 3 = (X-2)Q_2 - 4$$

$$\begin{cases} P(-1) = 3 \\ P(2) = -4 \end{cases}$$

d'autre part $P = (X+1)(X-2)Q_3 + aX + b$

$$\Leftrightarrow \begin{cases} P(-1) = -a + b \\ P(2) = 2a + b \end{cases}$$

$$\text{d'où } \begin{cases} -a + b = 3 \quad (1) \\ 2a + b = -4 \quad (2) \end{cases} \Leftrightarrow \begin{cases} (2) - (1) \Rightarrow 3a = -7 \\ \Rightarrow a = -\frac{7}{3} \\ (1) \Rightarrow b = 3 + a \\ = 3 - \frac{7}{3} = \frac{2}{3} \end{cases}$$

$$\text{d'où } R(x) = -\frac{7}{3}x + \frac{2}{3}$$

Ex2

$$P(x) = x^6 + 4x^5 + 5x^4 + x^3 - 4x^2 - 5x - 2$$

$$1) P(-1) = \cancel{1} - 4 + \cancel{5} - \cancel{1} - 4 + \cancel{5} - 2 = 10 - 10 = 0$$

$$P'(x) = 6x^5 + 20x^4 + 20x^3 + 3x^2 - 8x - 5$$

$$P'(0) = -6 + \cancel{20} - \cancel{20} + 3 + 8 - 5 = 11 - 11 = 0$$

$$P''(x) = 30x^4 + 80x^3 + 60x^2 + 6x - 8$$

$$P''(-1) = 30 - 80 + 60 - 6 - 8 = -4 \neq 0$$

donc (-1) est une racine double de P .

$$\begin{aligned} 2) P(j) &= j^6 + 4j^5 + 5j^4 + j^3 - 4j^2 - 5j - 2 \\ &= 1 + 4j^2 + 5j + 1 - 4j^2 - 5j - 2 = 0 \end{aligned}$$

donc j est une racine de P .

(2)

3) j est une racine de $P \in \mathbb{R}[X]$
donc $\bar{j}$ est une racine de P .

d'où $P = (X+1)^2(X-j)(X-\bar{j})Q$
cherchons Q .

$$\begin{aligned}(X+1)^2(X^2+X+1) &= (X^2+2X+1)(X^2+X+1) \\ &= X^4 + X^3 + X^2 + 2X^3 + 2X^2 + 2X + X + 1 \\ &= X^4 + 3X^3 + 4X^2 + 3X + 1\end{aligned}$$

$$\begin{array}{r|l} X^6 + 4X^5 + 5X^4 + X^3 - 4X^2 - 5X - 2 & X^4 + 3X^3 + 4X^2 + 3X + 1 \\ X^6 + 3X^5 + 4X^4 + 3X^3 + X^2 & X^2 + X - 2 \end{array}$$

$$X^5 + X^4 - 2X^3 - 5X^2 - 5X - 2$$

$$X^5 + 3X^4 + 4X^3 + 3X^2 + X$$

$$-2X^4 - 6X^3 - 8X^2 - 6X - 2$$

$$-2X^4 - 6X^3 - 8X^2 - 6X - 2$$

0

$$\Rightarrow P = (X+1)^2(X-j)(X-\bar{j})(X-1)(X+2)$$

$$\text{dans } \mathbb{R}[X] \quad P = (X+1)^2(X^2+X+1)(X-1)(X+2)$$

(2)

$$4) Q = X^4 - 1 = (X^2 - 1)(X^2 + 1) \\ = (X-1)(X+1)(X-i)(X+i)$$

dans $R[X]$, $Q = (X-1)(X+1)(X^2+1)$

$$\begin{aligned} \rightarrow F(X) &= \frac{Q(X)}{P(X)} = \frac{(X-1)(X+1)(X^2+1)}{(X+1)^2(X^2+X+1)(X-1)(X+2)} \\ &= \frac{X^2+1}{(X+1)(X+2)(X^2+X+1)} \\ &= \frac{a}{X+1} + \frac{b}{X+2} + \frac{cX+d}{X^2+X+1} \end{aligned}$$

$$(X+1)F \Big|_{X=-1} = 2 = a$$

$$(X+2)F \Big|_{X=-2} = \frac{-5}{3} = b$$

$$XF(X) \Big|_{X \rightarrow +\infty} = 0 = a + b + c$$

$$\Rightarrow c = -a - b = -2 + \frac{5}{3} = -\frac{1}{3}$$

$$F(0) = \frac{1}{2} = a + \frac{b}{2} + d$$

$$\begin{aligned} \Rightarrow d &= \frac{1}{2} - a - \frac{b}{2} = \frac{1}{2} - 2 + \frac{5}{6} \\ &= \frac{3 - 12 + 8}{6} = -\frac{2}{3} \end{aligned}$$

(4)

Exercice n°3.

$$F(x) = \frac{x^5 - 2x + 4}{(x-1)^2(x^2+x+1)}$$

1) Cherchons la partie entière (E)

$$(x-1)^2(x^2+x+1) = x^4 - x^3 - x + 1.$$

$$\begin{array}{r|l} x^5 - 2x + 4 & x^4 - x^3 - x + 1 \\ - (x^5 - x^4 - x^2 + x) & \\ \hline x^4 + x^2 - 3x + 4 & \\ - (x^4 - x^3 - x + 1) & \\ \hline x^3 + x^2 - 2x + 3 & \end{array}$$

$$\Rightarrow F(x) = x + 1 + \frac{x^3 + x^2 - 2x + 3}{(x-1)^2(x^2+x+1)}$$

on pose $H(x) = \frac{x^3 + x^2 - 2x + 3}{(x-1)^2(x^2+x+1)}$

P.E.S $H(x) = \frac{a}{x-1} + \frac{b}{(x-1)^2} + \frac{cx+d}{x^2+x+1}$

$$\lim_{x \rightarrow 1} (x-1)^2 H(x) = \frac{3}{3} = 1 = b$$

$$\lim_{x \rightarrow +\infty} x H(x) \Rightarrow a + c = 1$$

$$H(0) = 3 = -a + b + d$$

$$H(-1) = \frac{5}{4} = \frac{-a}{2} + \frac{b}{4} - c + d$$

$$\Rightarrow \begin{cases} a + c = 1 & \textcircled{1} \\ -a + 1 + d = 3 & \textcircled{2} \\ -\frac{a}{2} + \frac{1}{4} - c + d = \frac{5}{4} & \textcircled{3} \end{cases}$$

$$\Rightarrow \begin{cases} \textcircled{1} + \textcircled{3} \Rightarrow \frac{a}{2} + d = 0 & \textcircled{1}' \\ 2 \Rightarrow -a + d = 2 & \textcircled{2}' \end{cases}$$

$$\Rightarrow \begin{cases} \textcircled{1}' - \textcircled{2}' \Rightarrow \frac{3}{2}a = -2 \Rightarrow a = -\frac{4}{3} \\ \textcircled{2}' \Rightarrow d = 2 + a = 2 - \frac{4}{3} = \frac{2}{3} \end{cases}$$

$$\textcircled{1} \Rightarrow c = 1 - a = 1 + \frac{4}{3} = \frac{7}{3}$$

$$\Rightarrow \begin{cases} a = -\frac{4}{3} \\ b = 1 \\ c = \frac{7}{3} \\ d = \frac{2}{3} \end{cases}$$

P.E.S

$$= \frac{a}{x+1} + \frac{b}{x+2} + \frac{c}{x+3}$$

$$2) \quad F(x) = \frac{3x^2 + 12x + 11}{(x+1)(x+2)(x+3)}$$

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$$(x+1) F(x) \Big|_{x=-1} = \frac{2}{2} = \boxed{1 = a}$$

$$(x+2) F(x) \Big|_{x=-2} = \frac{-1}{-1} = \boxed{1 = b}$$

$$(x+3) F(x) \Big|_{x=-3} = \frac{2}{2} = \boxed{1 = c}$$

$$3) \quad F(x) = \frac{x^5 + 2x - 1}{x^4 - 1}$$

Check E:

$$\begin{array}{r|l} x^5 + 2x - 1 & x^4 - 1 \\ -x^5 - x & \\ \hline 3x - 1 & x \end{array}$$

$$\Rightarrow F = x + \frac{3x - 1}{(x-1)(x+1)(x^2+1)}$$

$$H(x) = \frac{3x - 1}{(x-1)(x+1)(x^2+1)} = \frac{a}{x-1} + \frac{b}{x+1} + \frac{cx+d}{x^2+1}$$

$$(x-1) \Big|_{x=1}^+ = \frac{1}{2} = a$$

$$(x+1) \Big|_{x=-1}^+ = 1 = b$$

$$(x^2+1) \Big|_{x=i}^+ = \frac{3i-1}{i^2-1} = ci+d$$

$$= -\frac{3}{2}i + \frac{1}{2} = ci+d$$

$$\Rightarrow \begin{cases} c = -\frac{3}{2} \\ d = \frac{1}{2} \end{cases}$$

$$\Rightarrow F = x + \frac{1}{2(x-1)} + \frac{1}{x+1} + \frac{-\frac{3}{2}x + \frac{1}{2}}{x^2+1}$$